

Crop Management OS

Argonaut and HMI

February 2026

CONFIDENTIAL

Patent Pending



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Hardware: Raspberry Pi 5 running Bookworm OS (future thinking Jetson Orin Nano with embedded horticultural AI)

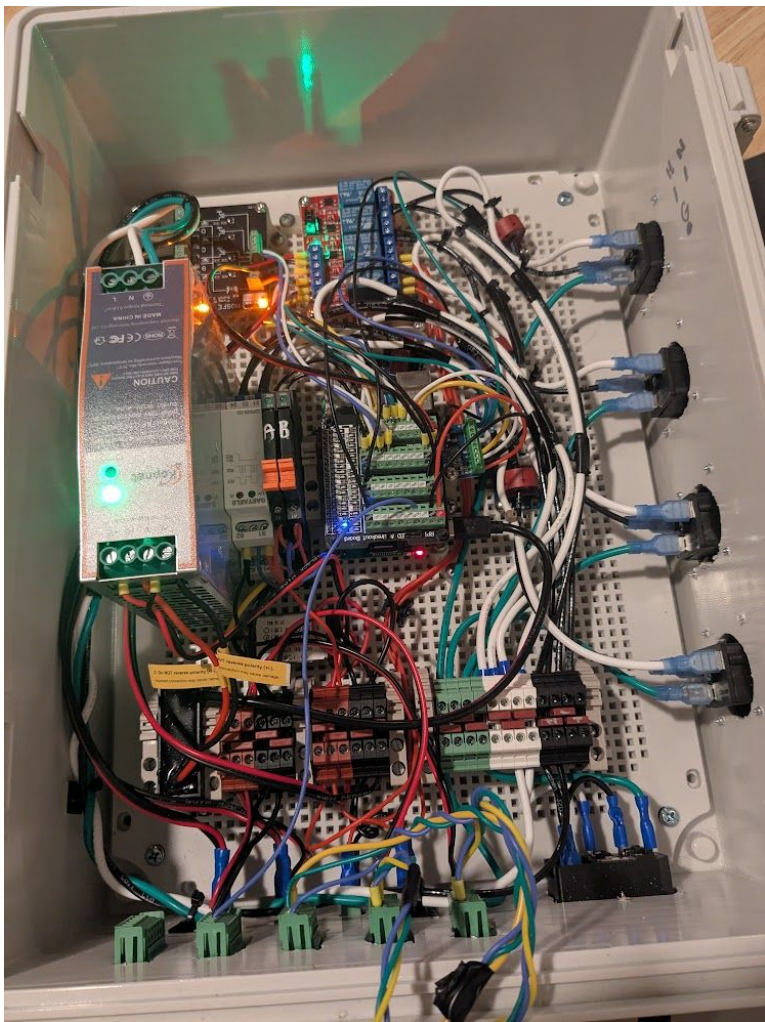
Argonaut software: Python

How to pass data/commands back & forth from Argonaut's software to the dashboard? How does REST API work?

- 1) Argonaut software can open files and inject commands (json?)
 - a) Example: dir/Lights.json user_request=on; override_automation=True; automation_enabled=False
 - b) Can your app create these files? Other files?
- 2) Data? Currently I'm storing data locally on the hardware in csv files with a utc datetime stamp
 - a) I can write a script to create whatever file you need to read.

Other notes: user manual controls = lights on/off, dosing pump, etc
User input (integers, toggles, text)

Any chance you're on Trello?





Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Opens to current information

Plot: pH gauge

Plot: EC gauge

Plot: VPD gauge

Plot: DLI gauge

Plot: plant performance gauge

Countdown: days til harvest

Running cost: \$

Plot: indicator lights on/off

Plot: indicator pump on/off

Plot: indicator fan on/off

Plot: indicator other on/off

Plot: timeseries of operations (lights, pumps, fans, dosing) past 48 hours.

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

What are you growing? (select dropdown)

How many plants? (enter integer)

What date did you seed? (enter
mm/dd/yyyy)

What date did you harvest? (enter
mm/dd/yyyy)

How much did you yield? (enter integer)
units (dropdown: ounces, grams, kg,
pounds)

Agree to share data with Cred Hydro? (y/n)

Submit button (can only submit when all
information is entered)

Dashboard layout

Left Bar

Current Information

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Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

LED on/off/auto toggle (note: there will be latency for sending this command)

Light sensor connected? Toggle (y/n)

Are your crops outside in a greenhouse? (y/n)

Does your crops use LED lights? (y/n)

If you don't have the light sensor connected, please estimate PPFD at the canopy. (enter integer)

Total Watts of LEDs (enter integer)

Dashboard layout

Left Bar

Current Information

Crop

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Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Pump on/off/auto toggle (note: there will be latency for sending this command)

How often do you need the pump running? (dropdown: continuously, periodically)

If periodically, enter frequency and duration (enter integer) units (dropdown: minutes, hours); (enter integer) minutes

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Fan on/off/auto toggle (note: there will be latency for sending this command)

How often do you need the fan running? (dropdown: continuously, periodically)

If periodically, enter frequency and duration (enter integer) units (dropdown: minutes, hours); (enter integer) minutes

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

X on/off/auto (note: there will be latency for sending this command)

How often do you need the X running?
(dropdown: continuously, periodically)

If periodically, enter frequency and duration (enter integer) units
(dropdown: minutes, hours); (enter integer) minutes

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Slider bar: designate desired pH range (Argonaut's machine learning alg will also suggest)

Display current pH (can be same indicator as in current information)
Display time of last dose

The user needs to train Argonaut to dose the reservoir with pH_up or pH_down. To do that, in the beginning the user needs to run the doser manually, and then Argonaut will remember the resulting change in pH. Over time, Argonaut will be more confident in running the doser itself.

Manually run pH up doser: (enter integer, units seconds) (Submit button)

Manually run pH down doser: (enter integer, units seconds) (Submit button)

Argonaut confidence level (1-10) pH up and pH down

Automatic on/off toggle

What time of day do you want Argonaut to run doser? (enter HH:MM AM/PM)

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Slider bar: designate desired EC range (Argonaut's machine learning alg will also suggest)

Display current EC (can be same indicator as in current information)
Display time of last dose

The user needs to train Argonaut to dose the reservoir with nutrient solution. To do that, in the beginning the user needs to run the doser manually, and then Argonaut will remember the resulting change in EC. Over time, Argonaut will be more confident in running the doser itself.

Manually run Nutrient doser: (enter integer, units seconds) (Submit button)

Argonaut confidence level (1-10) EC

Automatic on/off toggle

What time of day do you want Argonaut to run doser? (enter HH:MM AM/PM)

Dashboard layout

Left Bar

Current Information

Crop

Light

Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Display: pH probe last calibration

Display: EC probe last calibration

Self-calibrate pH: start button

- 1) Value low: (enter float) set button
- 2) 1-min pause
- 3) Value high: (enter float) set button
- 4) 1-min pause

Self-calibrate EC: start button

- 5) Value low: (enter float) set button
- 6) 1-min pause
- 7) Value high: (enter float) set button
- 8) 1-min pause

Dashboard layout

Left Bar

Current Information

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Pump

Fan

Other

pH Dosing

Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

View history of past harvests. User can select by harvest date and crop.

Display???

Dashboard layout

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Nutrient Dosing

Sensors

Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

Grower name: (text)

Company or institution: (text)

System type: (dropdown: nutrient film technique, buckets/deepwater culture, trays/shallow water culture)

Indoor or greenhouse or outside?

Volume Reservoir: (enter integer) units
Liters/gallons

Growers timezone: (drop down US and Canada timezones)

Farm location: (enter city,
state/province, country)

Dashboard layout

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Nutrient Dosing

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Harvests/Yields

Tell us about your system

Feedback to Credible Hydroponics

Main Pane

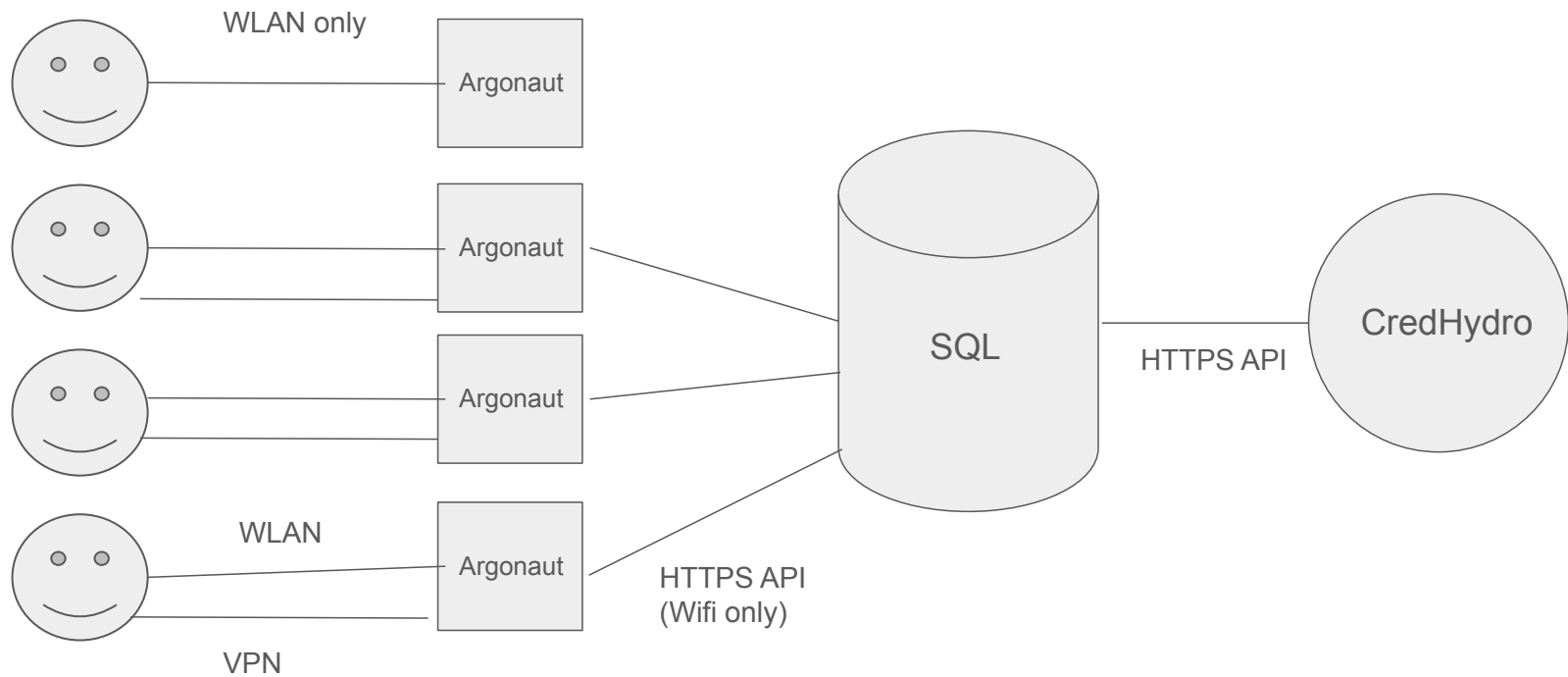
Feedback on hardware (text box)
submit button

Feedback on software (text box)
submit button

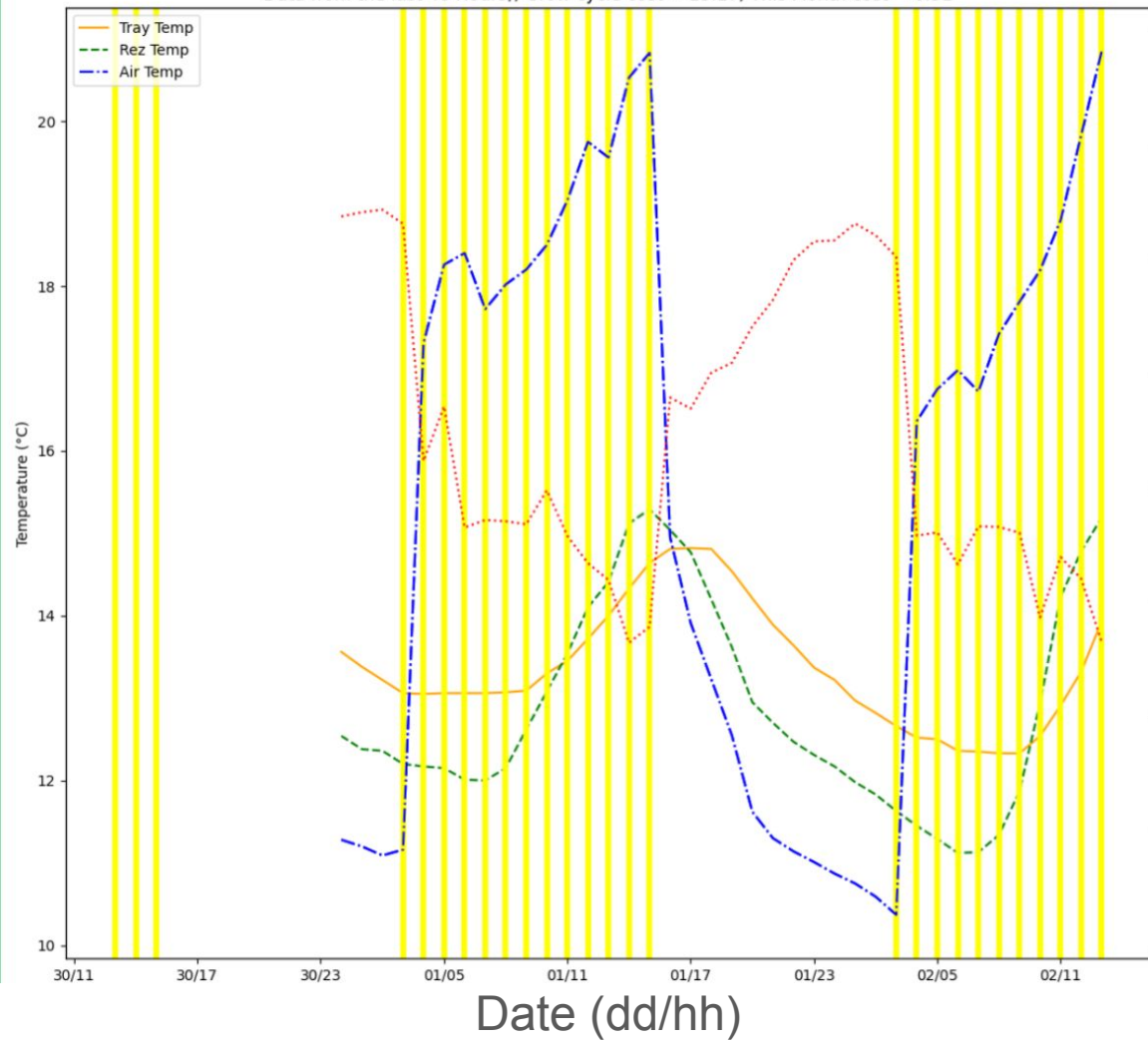
Users

Hardware

Cloud

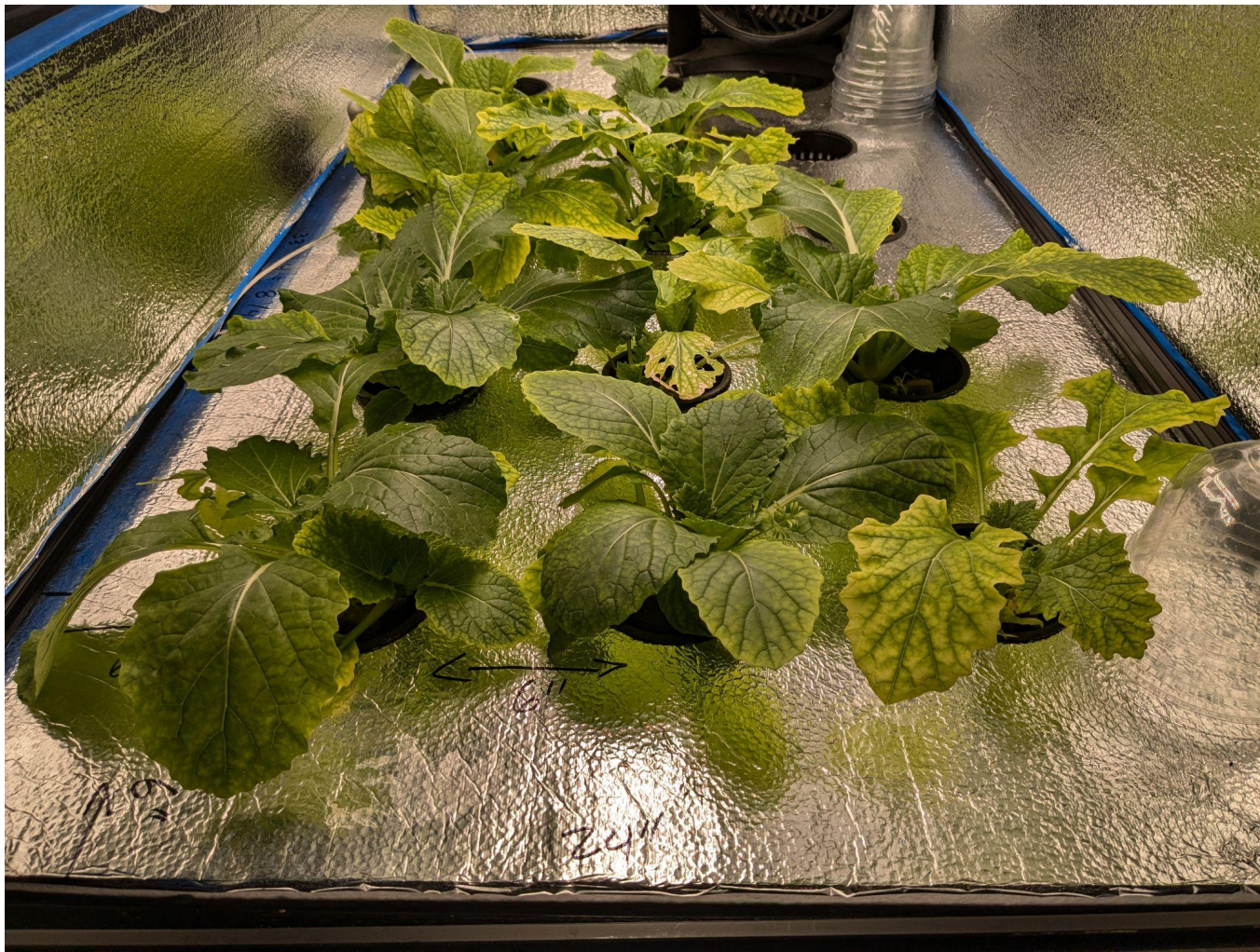


Data from the last 48 Hours); Grow cycle cost = 13.17; This Month cost = 0.51



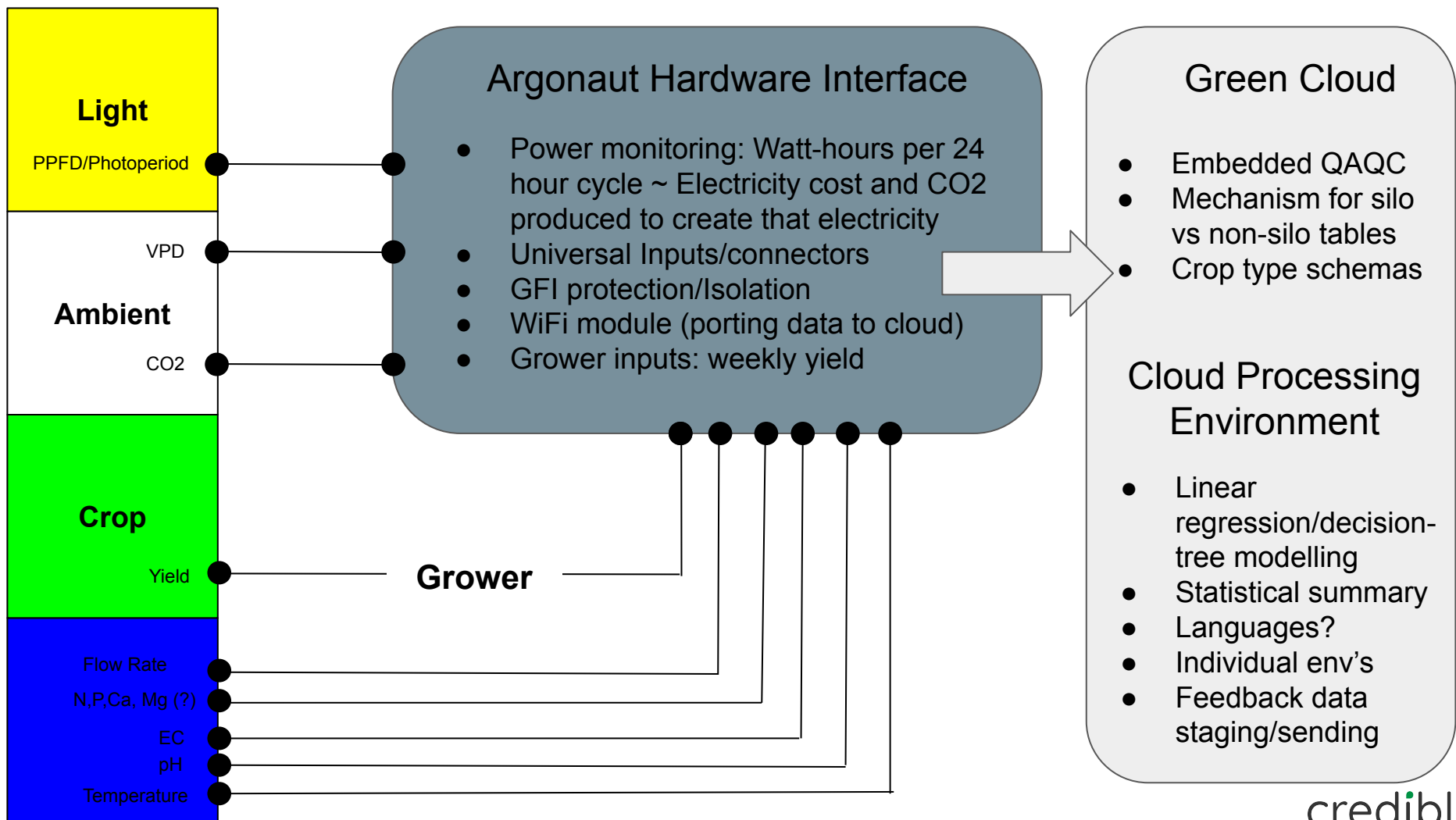
Vertical yellow bars in main plot are periods when lights are on

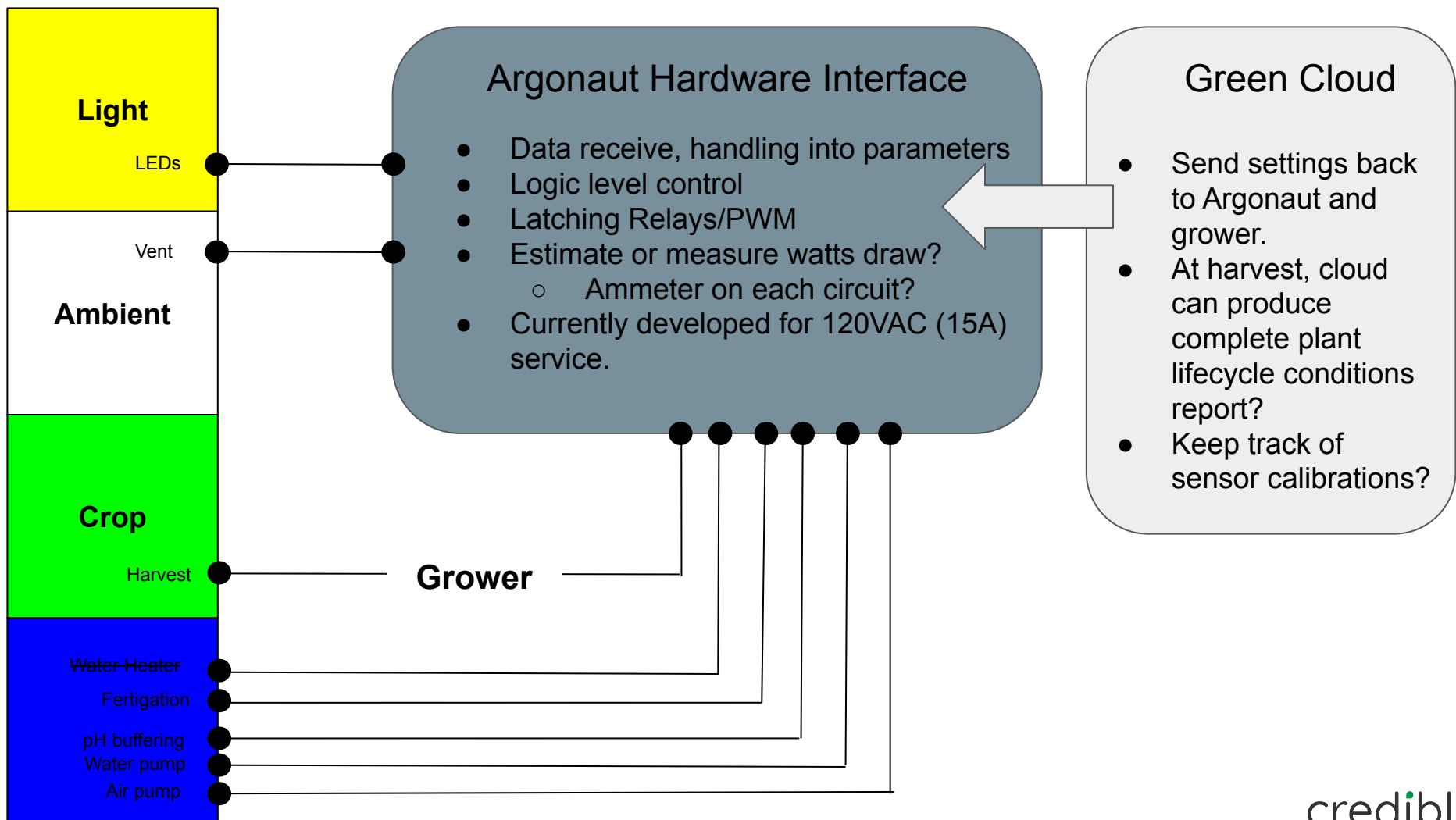
EC/pH: Red line is current. Yellow envelope is min/max range past 24 hours. Green box is ideal range.



Greens that I've been experimenting with in 2025.

On harvest, I collect quantity and mass of yield. Those data will serve as model outputs. The user will need to add harvest data into the app.



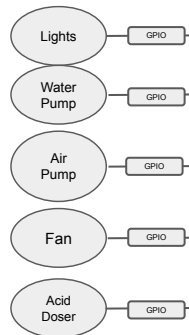


Hardware interfaces

Software Interfaces

Data

API



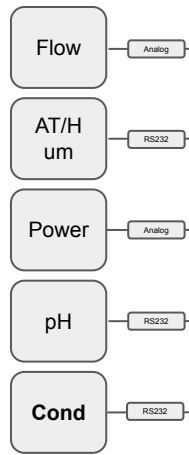
Functions

runDoser

runPump

runLights

Sensors



Sensors

read_data

energy_costs

write_data

plot

dli

ph_dosing_inputData

waterCirc_data

light_data

energy_data

caiso_status

ca_gridstatus

ambient_data

CA ISO

CO2

Sunlight

Energy rates

PLOT
OUTPUT

Data

Plant profiles

ph_dosing_inputData

waterCirc_data

light_data

energy_data

caiso_
status

dli

ambient_data

Feature Engineering

CO2 flux

Stomatal
Conductance

Vapor Pressure
Deficit

Emissions out

Electricity Cost

Future Social
Cost

Compile

Modelling

$f(\text{growth_index}) = \text{CO2_flux} \, dx/dt$

$f(\text{yield_cost}) = \text{sum}(\text{electricity_cost})/\text{unit}$

$f(\text{yield_mass_predict}) = \text{mass} += (\text{mass} * \text{growth_index})$

$f(\text{days_to_harvest}) = \text{user_input_harvest_mass_desired} - \text{yield_mass_predict} / \text{growth_index}$

Parameters to refine:
Growth_index, relative to
optimization parameters - staged
in the plant profiles

How to represent/quantify the trade-off
between cost and yield metrics - and
make that user adjustable

